

Calculus II

Lecture 05

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Fall 2024

Integration by Parts

(1/7)

Definition (1/4):

- The technique of *integration by parts* often makes it possible to reduce a complicated integral to a simpler integral. We know that if u and v are both differentiable functions, then uv is also differentiable and, by the product rule for derivatives,

$$\frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}.$$

Integration by Parts

(1/7)

Definition (2/4):

$$\frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}.$$

- This expression can be rewritten,

$$d(uv) = u \, dv + v \, du.$$

Integration by Parts (1/7)

Definition (3/4):

$$d(uv) = u \, dv + v \, du.$$

- Integrating both sides of this last equation gives

$$\int d(uv) = \int u \, dv + \int v \, du,$$
$$uv = \int u \, dv + \int v \, du.$$

Integration by Parts (1/7)

Definition (4/4):

$$uv = \int u \, dv + \int v \, du.$$

- Rearranging terms gives the following formula.

$$\int u \, dv = uv - \int v \, du$$

Integration by Parts (2/7)

Conditions for Integration by Parts

Integration by parts can be used only if the integrand satisfies the following conditions.

1. The integrand can be written as the product of two factors, u and dv .
2. It is possible to integrate dv to get v and to differentiate u to get du .
3. The integral $\int v du$ can be found.

$$\int u dv = uv - \int v du$$

Integration by Parts

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It is helpful to use the pattern:

The diagram illustrates the relationship between the variables in integration by parts. It shows two rows of equations: $u = \square$ and $dv = \square$ in the top row, and $du = \square$ and $v = \square$ in the bottom row. A green box labeled "Derivative" has a blue arrow pointing to the $u = \square$ equation and another blue arrow pointing to the $du = \square$ equation. A blue box labeled "Integral" has a blue arrow pointing to the $dv = \square$ equation and another blue arrow pointing to the $v = \square$ equation.

$$\begin{array}{ll} u = \square & dv = \square \\ du = \square & v = \square \end{array}$$

$$\int u \, dv = uv - \int v \, du$$

Integration by Parts

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GUIDELINES FOR INTEGRATION BY PARTS

1. Try letting dv be the most complicated portion of the integrand that fits a basic integration rule. Then u will be the remaining factor(s) of the integrand.
2. Try letting u be the portion of the integrand whose derivative is a function simpler than u . Then dv will be the remaining factor(s) of the integrand.

Note that dv always includes the dx of the original integrand.

$$\int u \, dv = uv - \int v \, du$$

Integration by Parts

(4/7)

Selecting u

An acronym that is very helpful to remember when using integration by parts is **LIATE**. Whichever function comes first in the following list should be u :

L	Logarithmic functions	$\ln(x)$, $\log_2(x)$, etc.
I	Inverse trig. functions	$\tan^{-1}(x)$, $\sin^{-1}(x)$, etc.
A	Algebraic functions	x , $3x^2$, $5x^2$, etc.
T	Trig. functions	$\cos(x)$, $\tan(x)$, etc.
E	Exponential functions	e^x , 2^x , etc.

$$\int u \, dv = uv - \int v \, du$$

Note that: there are exceptions to LIATE.

Example 1 (1/5):

Find $\int x e^x dx$.

INTEGRATION BY PARTS (5/7)

Integration by Parts

Example 1 (2/5):

$$\int \underbrace{(xe^x)}_u \underbrace{(dx)}_{dv}$$

$$\text{Find } \int xe^x dx.$$

There are several ways to do parts.

$$\int \underbrace{(x)}_u \underbrace{(e^x dx)}_{dv}, \quad \int \underbrace{(e^x)}_u \underbrace{(x dx)}_{dv}, \quad \int \underbrace{(1)}_u \underbrace{(xe^x dx)}_{dv},$$

Integration by Parts

Example 1 (2/5):

$$\int \underbrace{(xe^x)}_u \underbrace{(dx)}_{dv}$$

$$\text{Find } \int xe^x dx.$$

There are several ways to do parts.

$$\int \underbrace{(x)}_u \underbrace{(e^x dx)}_{dv},$$

$$\int \underbrace{(e^x)}_u \underbrace{(x dx)}_{dv},$$

$$\int \underbrace{(1)}_u \underbrace{(xe^x dx)}_{dv},$$

Integration by Parts

Example 1 (3/5):

$$\int \underbrace{(x)}_u \underbrace{(e^x dx)}_{dv}$$

$$\text{Find } \int x e^x dx.$$

There are several ways to do parts.

$$u = x$$

$$dv = e^x dx$$

$$du = dx$$

$$v = e^x$$

Integration by Parts

Example 1 (4/5):

$$\int \underbrace{(x)}_u \underbrace{(e^x dx)}_{dv}$$

$$\text{Find } \int x e^x dx.$$

There are several ways to do parts.

$$u = x$$

$$dv = e^x dx$$

$$du = dx$$

$$v = e^x$$

$$\int u dv = uv - \int v du$$

Integration by Parts

Example 1 (5/5):

$$\int \underbrace{(x)}_u \underbrace{(e^x dx)}_{dv}$$

$$\text{Find } \int x e^x dx.$$

There are several ways to do parts.

$$u = x$$

$$dv = e^x dx$$

$$du = dx$$

$$v = e^x$$

$$\begin{aligned} \int x e^x dx &= x e^x - \int e^x dx \\ &= x e^x - e^x + C. \end{aligned}$$

Example 2 (1/4):

Find $\int x \sin x \, dx$.

Example 2 (2/4):

Find $\int x \sin x \, dx$.

Let

$$u = x$$

$$dv = \sin x \, dx$$

$$du = dx$$

$$v = -\cos x$$

Example 2 (3/4):

$$\int \underbrace{x}_u \underbrace{\sin x \, dx}_{dv}$$

$$\begin{array}{ll} u = x & dv = \sin x \, dx \\ du = dx & v = -\cos x \end{array}$$

$$= \underbrace{x}_u \underbrace{(-\cos x)}_v - \int \underbrace{(-\cos x)}_v \underbrace{dx}_{du}$$

Example 2 (4/4):

$$\begin{array}{ll} u = x & dv = \sin x \, dx \\ du = dx & v = -\cos x \end{array}$$

$$= \overbrace{x}^u \overbrace{(-\cos x)}^v - \int \overbrace{(-\cos x)}^v \overbrace{dx}^{du}$$

$$= -x \cos x + \int \cos x \, dx$$

$$= -x \cos x + \sin x + C$$

Example 3 (1/4):

Find $\int x^2 \ln x \, dx$.

Integration by Parts

Example 3 (2/4):

Find $\int x^2 \ln x \, dx$.

$$u = \ln x$$

$$dv = x^2 \, dx$$

$$du = 1/x \, dx$$

$$v = x^3/3$$

Integration by Parts

Example 3 (3/4):

$$\int x^2 \ln x \, dx.$$

$$\int u \, dv = uv - \int v \, du$$

$$u = \ln x$$

$$dv = x^2 \, dx$$

$$du = 1/x \, dx$$

$$v = x^3/3$$

Integration by Parts

Example 3 (4/4):

$$\int x^2 \ln x \, dx.$$

$$u = \ln x$$

$$dv = x^2 \, dx$$

$$du = 1/x \, dx$$

$$v = x^3/3$$

$$\int u \, dv = uv - \int v \, du$$

$$= \frac{x^3}{3} \ln x - \int \left(\frac{x^3}{3} \right) \left(\frac{1}{x} \right) dx$$

$$= \frac{x^3}{3} \ln x - \frac{1}{3} \int x^2 \, dx$$

$$= \frac{x^3}{3} \ln x - \frac{x^3}{9} + C.$$

Integration by Parts

▣ **Example 4 (5/4):**

$$\int \tan^{-1} x \, dx$$

Let $u = \tan^{-1} x \implies du = \frac{1}{x^2+1} dx$

$$dv = dx \implies v = x$$

Hence

$$\int \tan^{-1} x \, dx = x \tan^{-1} x - \int \frac{x}{x^2 + 1} dx$$

$$= x \tan^{-1} x - \frac{1}{2} \ln(x^2 + 1)$$

Integration by Parts

▣ Example 5 (6/4):

$$\int \ln(x) dx$$

Let $u = \ln(x) \Rightarrow du = \frac{1}{x} dx$

$$dv = dx \Rightarrow v = x$$

Then

$$\int \ln(x) dx = x \ln(x) - \int x \frac{dx}{x}$$

$$= x \ln(x) - x$$

Integration by Parts

▣ **Example 6 (7/4):**

$$\int \sqrt{x^2 + a^2} \, dx$$

Let $u = \sqrt{x^2 + a^2} \Rightarrow du = \frac{x}{\sqrt{x^2 + a^2}}$

$$dv = dx \Rightarrow v = x$$

Hence

$$\int \sqrt{x^2 + a^2} \, dx = x \sqrt{x^2 + a^2} - \int \frac{x^2}{\sqrt{x^2 + a^2}} \, dx$$

$$I = x \sqrt{x^2 + a^2} - \int \frac{x^2 + a^2 - a^2}{\sqrt{x^2 + a^2}} \, dx$$

$$I = x \sqrt{x^2 + a^2} - \left(I - \int \frac{a^2}{\sqrt{x^2 + a^2}} dx \right)$$

$$I = x \sqrt{x^2 + a^2} - I + a^2 \int \frac{dx}{\sqrt{x^2 + a^2}}$$

$$2I = x \sqrt{x^2 + a^2} + a^2 \sinh^{-1} \left(\frac{x}{a} \right)$$

$$I = \frac{1}{2} x \sqrt{x^2 + a^2} + \frac{1}{2} a^2 \sinh^{-1} \left(\frac{x}{a} \right)$$

Integration by Parts

▣ **Example 7 (8/4):**

$$\int e^{3x} \cos 2x \, dx$$

Put $\int e^{3x} \cos 2x \, dx = I$

Let $u = e^{3x} \Rightarrow du = 3e^{3x} \, dx$

$$dv = \cos 2x \, dx \Rightarrow v = \frac{1}{2} \sin 2x$$

$$I = \frac{1}{2} e^{3x} \sin 2x - \frac{3}{2} \int e^{3x} \sin 2x \, dx$$

Integrating $\int e^{3x} \sin 2x \, dx$ by parts again we obtain :

Let $u = e^{3x} \Rightarrow du = 3e^{3x} \, dx$

$$dv = \sin 2x \Rightarrow v = -\frac{1}{2} \cos 2x$$

$$I = \frac{1}{2} e^{3x} \sin 2x - \frac{3}{2} \left[-\frac{1}{2} e^{3x} \cos 2x + \frac{3}{2} \int e^{3x} \cos 2x \, dx \right]$$

$$I = \frac{1}{2} e^{3x} \sin 2x + \frac{3}{4} e^{3x} \cos 2x - \frac{9}{4} I$$

$$\left(1 + \frac{9}{4} \right) I = \frac{1}{2} e^{3x} \sin 2x + \frac{3}{4} e^{3x} \cos 2x$$

$$I = \frac{e^{3x}}{13} [2 \sin 2x + 3 \cos 2x]$$

Similarly :

$$\int e^{3x} \cos 2x \, dx = \frac{e^{3x}}{13} [2 \sin 2x + 3 \cos 2x]$$

Thank You

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